Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller

Interface Design Description

Reference	TBD **[TBD-1]**
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Document change record

Issue	Rev.	Date	Change
1	0	2026-10-12	First draft

1.0 Purpose and Scope

This document defines each interface identified in the architecture [2, Sec. 7.0]: the format and content of every data element that crosses it, its communication method and protocol, and the physical connections between the two boards. It covers the minimum content of an interface design description [1, Sec. 1].

2.0 Documents

[1] NASA, *NASA Software Engineering Handbook*, NASA-HDBK-2203, Version D, Topic 5.02 "IDD – Interface Design Description".
[2] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Architecture*, Issue 1, Revision 0, Draft.
[3] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Requirements Specification*, Issue 1, Revision 0, Draft.
[4] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Hazard Analysis*, Issue 1, Revision 1, Draft.
[5] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Concept of Operations*, Issue 1, Revision 1, Draft.
[6] S. Josefsson and I. Liusvaara, *Edwards-Curve Digital Signature Algorithm (EdDSA)*, RFC 8032, January 2017.
[7] NIST, *Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC*, SP 800-38D, November 2007.
[8] Arm, *ARM Debug Interface v5 Architecture Specification*, IHI 0031A, 2006.
[9] STMicroelectronics, *STM32G4 Series Reference Manual*, RM0440 Rev 9.
[10] STMicroelectronics, *How to Use the USART Protocol in the Bootloader on STM32 MCUs*, AN3155 Rev 22.
[11] STMicroelectronics, *Introduction to System Memory Boot Mode on STM32 MCUs*, AN2606 Rev 70.
[12] STMicroelectronics, *Discovery Kit with STM32MP257F MPU*, UM3385 Rev 2.

3.0 Overview

Figure 1 shows every interface and the signals it uses. Table 1 gives each interface's type, priority and use, and whether it is safety-critical [1, Sec. 1].

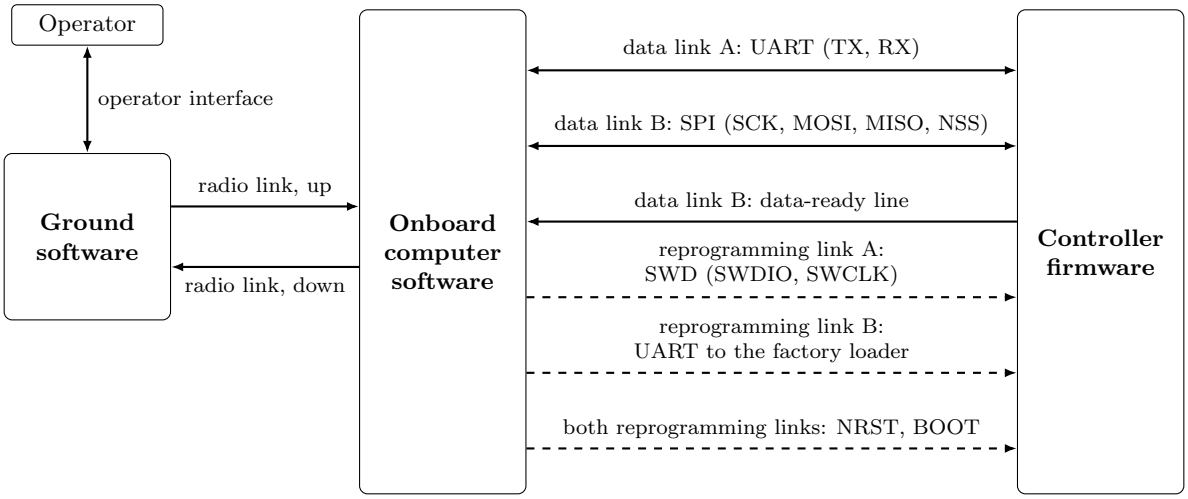


Figure 1: The interfaces and their signals. Solid: used in operation. Dashed: used only to restore or reload the controller.

Interface	Type	Priority	Used	Safety-critical
Operator interface	Commands in, state out	ONLY	ALWAYS	YES
Radio link	Data transfer, encrypted and signed	ONLY	AT CONTACTS	YES
Data link A	Data transfer	PRIMARY	ALWAYS	YES
Data link B	Data transfer	BACKUP	ALWAYS	YES
Reprogramming link A	Memory load	PRIMARY	RESTORE	YES
Reprogramming link B	Memory load	BACKUP	RESTORE	YES

Table 1: Interface types [**TBR-1**].

Legend. Priority: ONLY, the one interface for its traffic; PRIMARY, used first; BACKUP, used when the primary one is lost. Used: ALWAYS, whenever the System runs; AT CONTACTS, during contact windows; RESTORE, during a restore or a reload through the reprogramming link.

4.0 Message Layer

Every command, answer and report travels in the same frame on the radio link and on both data links (Figure 2). Multi-byte fields are little-endian. The check value is the CRC-32 of every byte before it, with the polynomial 0x4C11DB7 [9, p. 454]. A receiver drops a frame whose check value or version it does not accept [**TBR-2**].

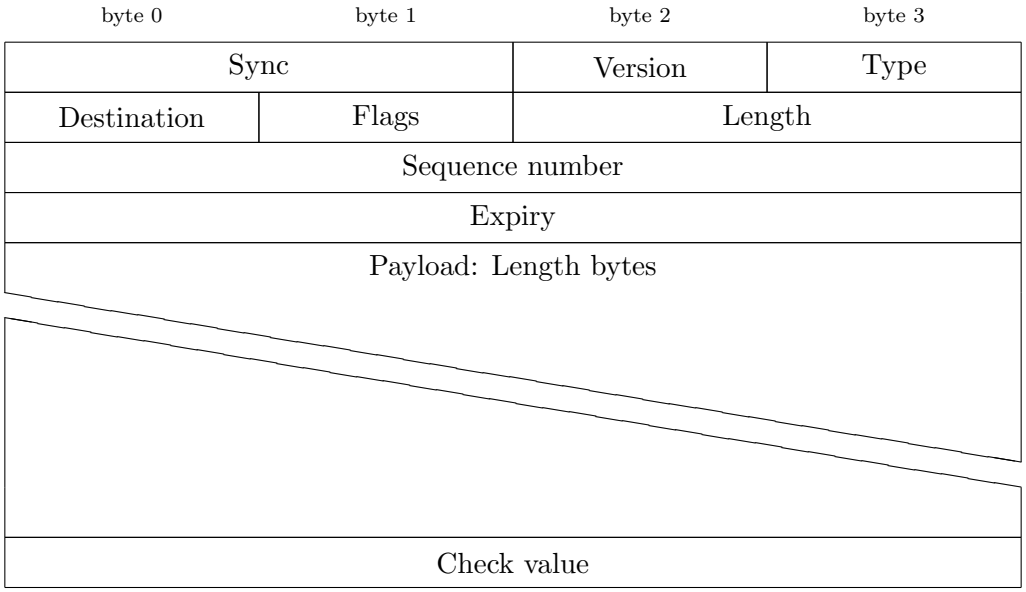


Figure 2: The frame. Each row is four bytes. The header is the first 16 bytes.

Field	Bytes	Values	Meaning
Sync	2	0xA5, 0x5A	Start of a frame
Version	1	0x01	Frame format version
Type	1	0x01 command, 0x02 answer, 0x03 report	What the payload holds
Destination	1	0x01 onboard computer, 0x02 controller, 0x03 ground	Who acts on the frame
Flags	1	bit 0	Set when the onboard computer releases a scheduled command; other bits 0
Length	2	0 to 1024	Payload length in bytes
Sequence number	4	1 to $2^{32} - 1$	One higher than the sender's last frame [3, REQ-065]
Expiry	4	seconds of ground time	After it, the command is refused [3, REQ-066, REQ-133]; 0 in answers and reports
Payload	Length		The command, answer or report
Check value	4	CRC-32	Proves the frame undamaged [3, REQ-107]

4.1 Radio Envelope

On the radio link each frame travels inside an envelope (Figure 3). The frame is signed twice with Ed25519, each signature 64 bytes [6, Sec. 1]. The frame and both signatures are then encrypted with AES-256 in Galois/Counter Mode, with a 12-byte nonce and a 16-byte tag [7, Secs. 5.2.1.1–5.2.1.2, pp. 8–9] **[TBR-3]**. The link encryptor builds the envelope. The onboard computer opens it, checks both signatures, and acts only when all three checks pass [3, REQ-127 to REQ-129].

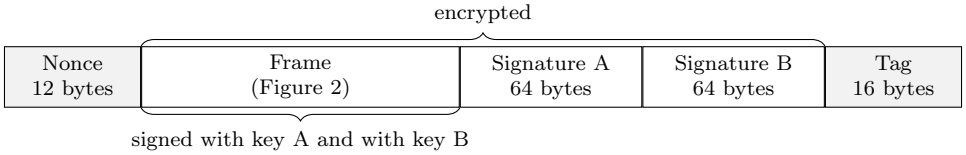


Figure 3: The radio envelope. The tag proves the encrypted part unchanged and encrypted with the right key; the signatures prove the frame comes from the operator.

4.2 Answers

Every command gets two answers [3, REQ-033]: RECEIVED when it passes the checks on arrival, then DONE with its result or REFUSED with a reason (Figure 4). A command sent again with a sequence number already run is answered ALREADY DONE and is not run twice [3, REQ-031]. An envelope that fails any of its checks is dropped and logged without an answer, so that a sender without the keys learns nothing [3, REQ-064] [**TBR-4**].

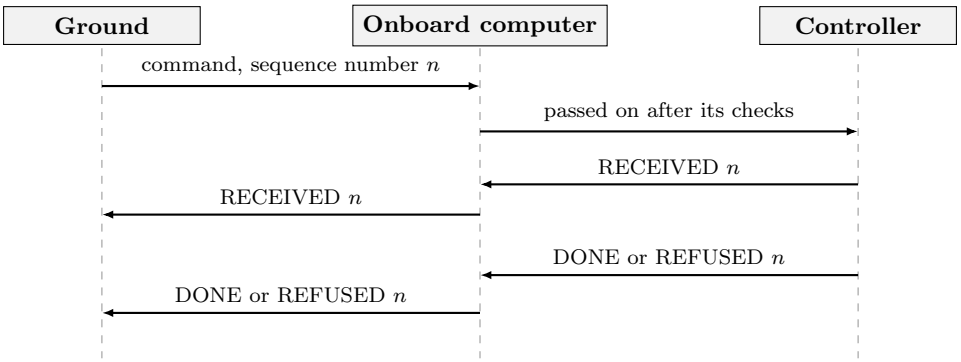


Figure 4: A command to the controller and its two answers. DONE carries the result, REFUSED the reason. A command to the onboard computer is answered by the onboard computer.

An answer's payload is six bytes: the sequence number answered (4 bytes); the kind (1 byte): 0x01 RECEIVED, 0x02 DONE, 0x03 REFUSED, 0x04 ALREADY DONE; and a code (1 byte): for DONE, 0x00 on success; for REFUSED, the reason below. The ground and the onboard software share this list [3, REQ-100].

Code	Reason	Requirement
0x01	Dangerous command without a valid arming command	REQ-029, REQ-112
0x02	Arming command expired	REQ-030
0x03	Command past its expiry	REQ-066, REQ-133
0x04	Old command sent again	REQ-065
0x05	Not allowed in the current mode	REQ-027
0x06	Impossible values, or an unknown command	REQ-028
0x07	No new firmware during a trial	REQ-050
0x08	Image damaged	REQ-005, REQ-044
0x09	Image not from the operator	REQ-006, REQ-043
0x0A	Image older than the minimum version	REQ-045
0x0B	Image built for another controller	REQ-046
0x0C	The controller's clock disagrees with the scheduled time	REQ-134
0x0D	A dangerous command cannot be scheduled	[**TBR-7**]

5.0 Commands, Reports and Firmware Images

5.1 Commands

A command's payload is its one-byte identifier followed by its fields. Identifiers 0x01 to 0x1F are commands to the onboard computer, 0x20 to 0x3F commands to the controller.

ID	Command	Fields (bytes)	Modes	Dangerous	Requirement
0x01	UPLOAD BEGIN	image number (2), image length (4)	ANY	NO	REQ-049
0x02	UPLOAD CHUNK	image number (2), offset (4), data (up to 256)	ANY	NO	REQ-049
0x03	UPLOAD END	image number (2)	ANY	NO	REQ-043 to 047
0x04	UPLOAD CANCEL	image number (2)	ANY	NO	REQ-051
0x05	DELIVER	image number (2)	OPERATE, FALLBACK	NO	REQ-048, REQ-050
0x06	SET LIMITS	minimum version (4), controller identity (2)	ANY	YES	REQ-053
0x07	RESTORE	none	ANY	YES	REQ-055
0x08	REPROGRAM	image number (2), region (1): bootloader, main firmware or last-resort firmware	ANY	YES	REQ-058, REQ-121
0x09	SET ONBOARD TIME	ground time (4)	ANY	YES	REQ-073
0x0A	SCHEDULE	run time (4), one complete frame	ANY	NO	REQ-068
0x0B	SEND REPORTS	first report number (4)	ANY	NO	REQ-074
0x0C	REPLACE KEY	which key (1): signing A, signing B or encryption; new key (32)	ANY	YES	REQ-084 to 086
0x0D	ARM	command identifier (1), its sequence number (4)	ANY	NO	REQ-029, REQ-112
0x20	SWITCH	none	OPERATE, FALLBACK	YES	REQ-021
0x21	RETURN TO PREVIOUS	none	OPERATE, TRIAL	YES	REQ-113
0x22	START FAILED	which copy (1)	OPERATE, FALLBACK	YES	REQ-019
0x23	LEAVE FALLBACK	none	FALLBACK	NO	REQ-117
0x24	SET CONTROLLER TIME	ground time (4)	OPERATE, TRIAL, FALLBACK	YES	REQ-132
0x25	GET STATUS	none	OPERATE, TRIAL, FALLBACK	NO	REQ-081
0x40	EQUIPMENT	set by the controlled equipment	[**TBD-2**]	[**TBD-2**]	—

Legend. Modes: the controller modes in which the command is accepted, within those that accept commands [5, Sec. 3.4]; ANY, every mode, RECOVERY included. Dangerous: YES, it needs an arming command first [3, REQ-029, REQ-112]; NO, it does not [**TBR-5**].

5.2 Arming

An ARM command names the one dangerous command it allows, by identifier and sequence number [**TBR-6**]. It allows no other command, and it expires after [**TBD-3**] [3, REQ-030]. The onboard computer and the controller each check it [3, REQ-029, REQ-112].

5.3 Scheduled Commands

SCHEDULE holds a complete frame on the onboard computer until its run time [3, REQ-068]. At that time the onboard computer checks the frame again against its check value [3, REQ-136] and passes it to the controller with Flags bit 0 set. The controller runs it only if its own clock agrees with the run time within **[TBD-4]** [3, REQ-134]. A dangerous command cannot be scheduled, because no operator is present at run time to arm it **[TBR-7]**.

5.4 Reports

The controller sends a HEALTH report at **[TBD-5]** and an EVENT report as soon as an event happens [3, REQ-037]. The onboard computer adds the report number and its own clock time before storing the report [3, REQ-070, REQ-071, REQ-135].

Report	Field	Bytes	Set by
ALL	report number	4	onboard computer
ALL	onboard computer clock time	4	onboard computer
ALL	controller clock time	4	controller
ALL	report kind: 0x01 HEALTH, 0x02 EVENT	1	controller
HEALTH	mode: 0x01 OPERATE, 0x02 TRIAL, 0x03 FALLBACK	1	controller
HEALTH	running firmware: kind (1), version (4)	5	controller
HEALTH	failed starts of the new copy	1	controller
HEALTH	trial time left, in seconds	2	controller
HEALTH	last self-test: 0x00 PASSED, 0x01 FAILED	1	controller
EVENT	event code	2	controller or onboard computer
EVENT	detail	up to 32	controller or onboard computer

Legend. Report: ALL, every report; HEALTH or EVENT, that kind only.

The event codes:

Code	Event	Sent by	Requirement
0x0001	fault detected	controller	REQ-038
0x0002	fall-back to the previous copy	controller	REQ-040
0x0003	fall-back to the last-resort firmware	controller	REQ-040
0x0004	self-test failed	controller	REQ-039
0x0005	change in memory detected	controller	REQ-041
0x0006	trial started	controller	REQ-037
0x0007	firmware confirmed	controller	REQ-037
0x0008	firmware given up	controller	REQ-037
0x0009	restart	controller	REQ-037
0x0101	data link lost	onboard computer	REQ-081
0x0102	data link working again	onboard computer	REQ-082
0x0103	restore done; detail: its result	onboard computer	REQ-060
0x0104	health reports dropped; detail: count and time span	onboard computer	REQ-076
0x0105	onboard computer clock differs from ground time	onboard computer	REQ-072

5.5 Firmware Images

Every firmware image, of every kind, has the same layout (Figures 5 and 6). Both signatures cover the header and the body; the controller checks them before it runs the image [3, REQ-005, REQ-006]. Images move in chunks of 256 bytes, the most the factory loader writes at once [10, p. 7], so one chunk size serves every path.

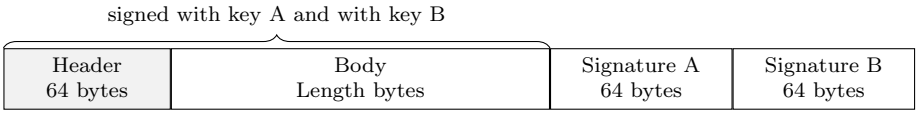


Figure 5: A firmware image.

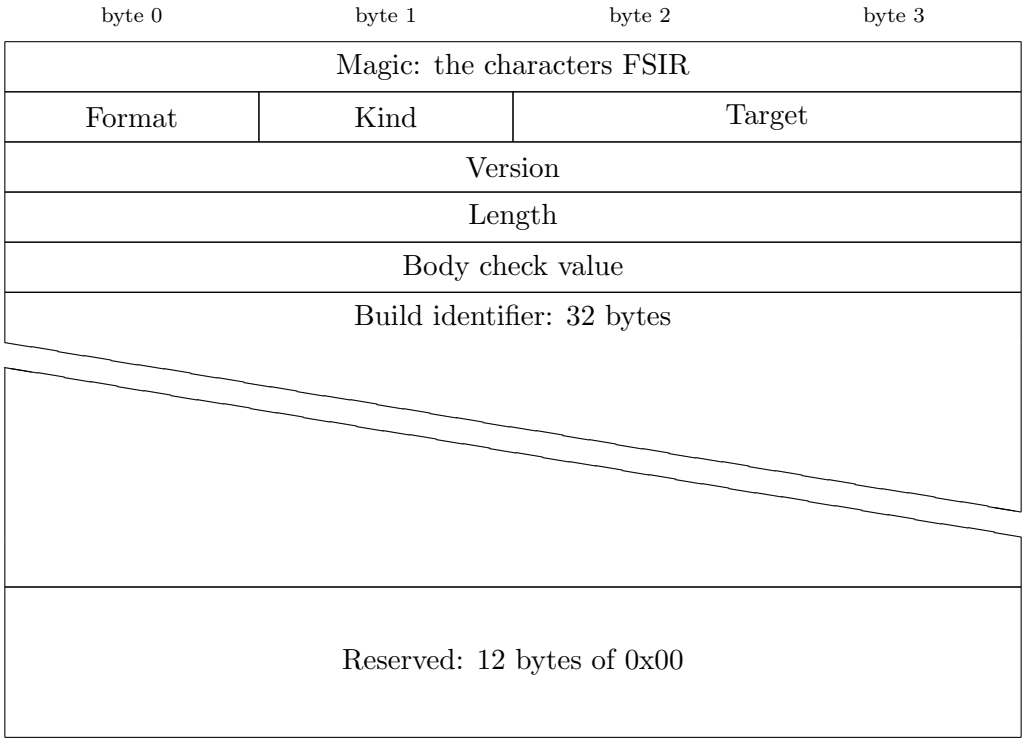


Figure 6: The image header. Kind: 0x01 bootloader, 0x02 main firmware, 0x03 last-resort firmware. Target: the controller identity [3, REQ-046]. Body check value: CRC-32 of the body. Build identifier: names the build that was tested [3, REQ-090].

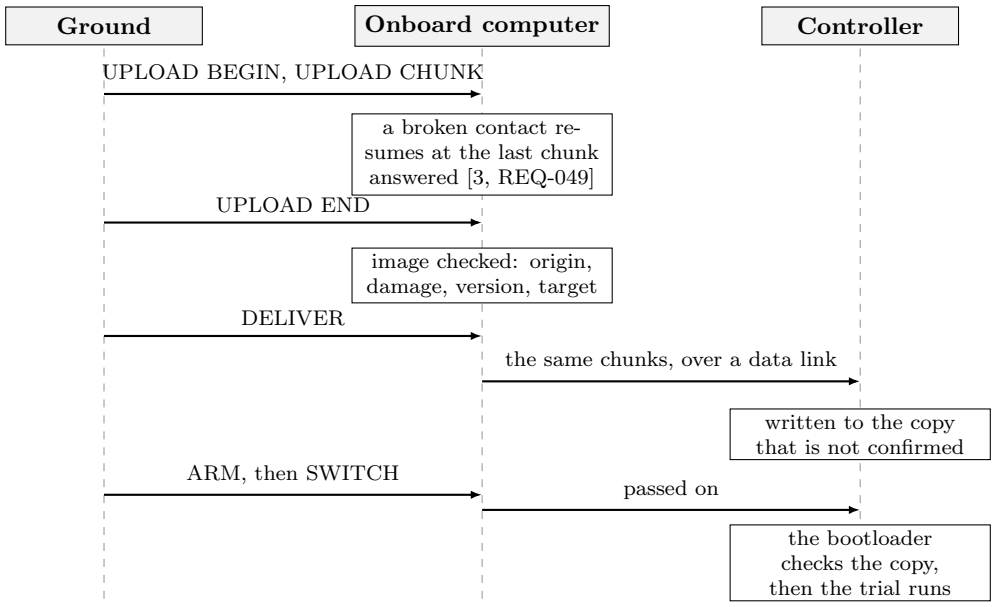


Figure 7: An update, as commands. Each command gets its two answers (Figure 4).

6.0 Links

6.1 Radio Link

On the bench, UDP over Ethernet between the ground computer and the onboard computer board [12, p. 2] stands in for the radio, one envelope per datagram **[TBR-8]**.

6.2 Data Link A: UART

115 200 baud, 8 data bits, no parity, 1 stop bit; a frame with a gap of more than 10 ms inside it is dropped **[TBR-9]**. On each data link the onboard computer sends GET STATUS every **[TBD-6]**. A link that misses **[TBD-7]** answers in a row is reported lost [3, REQ-081] and is used again once it answers [3, REQ-082].

6.3 Data Link B: SPI

The onboard computer drives the clock: SPI mode 0, 1 MHz **[TBR-10]**. It sends a frame by clocking it out with NSS low. The controller cannot start a transfer, so when it has a frame to send it raises the data-ready line; the onboard computer then clocks the 16 header bytes, reads Length, and clocks the rest (Figure 8). The data-ready line adds one pin on each board to the hardware needs [2, Sec. 8.0].

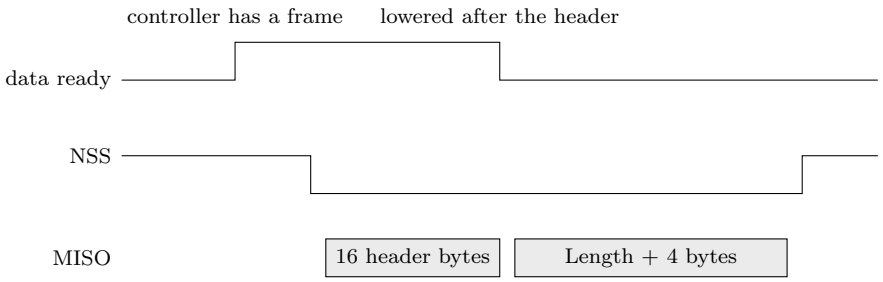


Figure 8: The controller sending a frame over SPI.

6.4 Reprogramming Link A: SWD

The onboard computer drives SWD from two general-purpose pins, with the controller held stopped, so that no controller software runs (Figure 9).

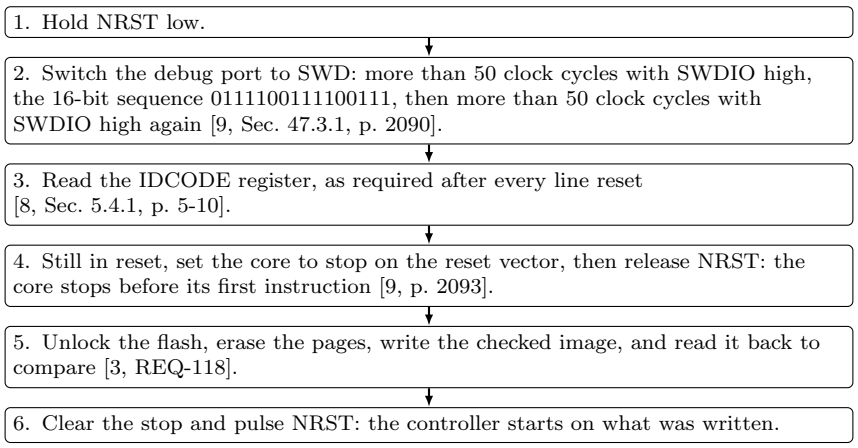


Figure 9: A restore through SWD.

6.5 Reprogramming Link B: Factory Loader

With the option bits as shipped (BOOT_LOCK = 0, nBOOT1 = 1, nSWBOOT0 = 1), the controller starts its factory loader when BOOT is high at reset [11, Table 2, p. 36]. The loader's protocol is fixed by its maker (Figure 10): 8 data bits, even parity, 1 stop bit [11, Table 115, p. 274]; the host sends 0x7F and the loader answers ACK, 0x79 [10, p. 5]; every command byte is followed by its complement, every block ends with the XOR of its bytes, and each packet is answered ACK 0x79 or NACK 0x1F [10, p. 8].

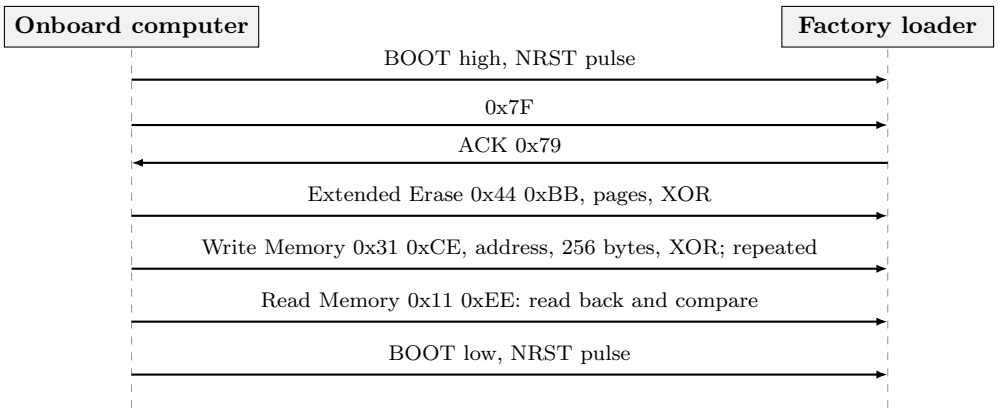


Figure 10: A restore through the factory loader. Each command is answered ACK before the next one. Command codes from [10, p. 7].

The onboard computer never sends the loader's Readout Protect (0x82) or Write Protect (0x63) commands [10, p. 7]: either would lock a way back in [3, REQ-125].

6.6 Operator Interface

The operator console shows every command with its state (sent, RECEIVED, DONE or REFUSED with the reason), every command still waiting for an answer [3, REQ-094], which firmware runs [3, REQ-095], and every report received. It accepts commands, planned sequences

of commands [3, REQ-115] and firmware releases. Signing with key B needs the operator's hardware security key [2, Sec. 5.4].

7.0 Wiring

The pin-to-pin wiring between the two boards is set in this section **[TBD-8]**.

8.0 Safety Features

Each feature of the interfaces that guards against a hazard of the hazard analysis [4, Sec. 4.0].

Feature	Interfaces	Hazards	Requirements
Check value on every frame and image body	radio link, data links	H-7, H-9	REQ-044, REQ-107
Encryption and two signatures	radio link	H-2, H-5	REQ-127 to 129
Sequence numbers	radio link, data links	H-6	REQ-031, REQ-065
Expiry in every command	radio link, data links	H-6	REQ-066, REQ-133
Arming before a dangerous command	radio link, data links	H-6	REQ-029, REQ-112
Two answers to every command	radio link, data links	H-11	REQ-033
Failed envelopes dropped and logged	radio link	H-5	REQ-064
Link check on each data link	data links	H-4	REQ-081
Only checked images written	reprogramming links	H-2, H-7	REQ-118
No protection command sent to the loader	reprogramming link B	H-1	REQ-125

A Acronyms

ACK Acknowledge
AES Advanced Encryption Standard
CRC Cyclic redundancy check
GCM Galois/Counter Mode
GPIO General-purpose input/output
MISO SPI line from the controller to the onboard computer
MOSI SPI line from the onboard computer to the controller
NACK Not acknowledge
NRST Reset pin
NSS SPI select line
SCK SPI clock line
SPI Serial Peripheral Interface
SWCLK SWD clock line
SWD Serial Wire Debug
SWDIO SWD data line
TBD To Be Determined
TBR To Be Resolved
UART Universal asynchronous receiver-transmitter
UDP User Datagram Protocol

B Traceability

Each requirement on an interface, and the section that meets it.

Requirement	Section
REQ-064 to 066, REQ-107	4.0 Message Layer
REQ-127 to 129	4.1 Radio Envelope

Requirement	Section
REQ-027, REQ-028, REQ-031, REQ-033 to 035, REQ-100	4.2 Answers
REQ-019, REQ-021, REQ-053, REQ-055, REQ-058, REQ-073, REQ-074, REQ-084 to 086, REQ-113, REQ-117, REQ-121, REQ-132	5.1 Commands
REQ-029, REQ-030, REQ-112	5.2 Arming
REQ-068, REQ-069, REQ-134, REQ-136	5.3 Scheduled Commands
REQ-037 to 041, REQ-060, REQ-070 to 072, REQ-076, REQ-135	5.4 Reports
REQ-005, REQ-006, REQ-043 to 051, REQ-090	5.5 Firmware Images
REQ-080 to 082, REQ-138	6.2, 6.3 Data Links
REQ-083, REQ-118, REQ-122, REQ-125	6.4, 6.5 Reprogramming Links
REQ-094, REQ-095, REQ-115	6.6 Operator Interface

C Open Items

The values below are not yet known.

TBD-1 Project name and document reference.

TBD-2 Commands to the controlled equipment, the modes that accept them, and which are dangerous; the equipment is outside this project.

TBD-3 Time after which an arming command expires [3, REQ-030].

TBD-4 Largest allowed difference between the controller's clock and a scheduled run time [3, REQ-134].

TBD-5 Rate of regular health reports [3, REQ-037].

TBD-6 Interval of the link check on each data link.

TBD-7 Number of missed answers in a row after which a data link is reported lost.

TBD-8 Pin-to-pin wiring between the two boards.

The proposals below await the author's acceptance.

TBR-1 Data link A and reprogramming link A used first; the B links used when the A links are lost (Section 3.0).

TBR-2 The frame layout, byte order and check value (Section 4.0).

TBR-3 Ed25519 signatures and AES-256 in Galois/Counter Mode (Section 4.1).

TBR-4 A failed envelope gets no answer (Section 4.2).

TBR-5 Which commands are dangerous (Section 5.1).

TBR-6 An arming command allows one named command only (Section 5.2).

TBR-7 A dangerous command cannot be scheduled (Section 5.3).

TBR-8 UDP over Ethernet as the bench stand-in for the radio (Section 6.1).

TBR-9 The UART settings of data link A (Section 6.2).

TBR-10 The SPI settings of data link B, and its data-ready line (Section 6.3).